

Leveraging AI in recruitment: enhancing intellectual capital through resource-based view and dynamic capability framework

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Abstract

Purpose – The rapid evolution of artificial intelligence (AI) is revolutionizing organizational operations and altering competitive landscapes. This study examines the influence of organizational resources on AI adoption in recruitment, focusing on their role in achieving competitive advantage through effective implementation.

Design/methodology/approach – This research utilizes a cross-sectional quantitative approach, applying partial least squares structural equation modeling (PLS-SEM) to data from 290 human resource (HR) professionals. It is grounded in the resource-based view (RBV) and dynamic capability framework (DCF).

Findings – The results reveal that HR competencies and open innovation significantly influence dynamic capabilities, which are essential for AI integration, supported by financial support and information technology (IT) infrastructure. These capabilities enable effective AI adoption, leading to a competitive advantage.

Research limitations/implications – The cross-sectional data in this study captures the current landscape of AI adoption in recruitment, providing a snapshot of the present scenario in a rapidly evolving technological environment.

Practical implications – This study offers HR professionals and managers strategic guidance on effectively integrating AI into recruitment processes. By enhancing HR competencies, fostering collaboration and ensuring sufficient financial and infrastructural support, organizations can navigate AI adoption challenges and secure a competitive advantage in a rapidly evolving technological landscape.

Social implications – The adoption of AI in recruitment can reduce biases, enhance diversity and improve fairness through standardized assessments. However, as AI technologies evolve, continuous human oversight is essential to ensure ethical use and to modify AI systems as needed, further reducing biases and addressing societal concerns in AI-driven recruitment processes.

Originality/value – This research introduces a novel framework that underscores the importance of integrating human expertise with advanced technological tools to ensure successful AI implementation. A key contribution is that HR professionals not only facilitate AI integration but also ensure accuracy, accountability and configure the most suitable AI tools for recruitment by collaborating with AI developers to meet the specific needs of the organization.

Keywords Recruitment, AI adoption, Artificial intelligence, Dynamic managerial capability, Competitive advantage, Resource-based view

Paper type Research paper

1. Introduction

AI is transforming recruitment by enhancing efficiency and competitive advantage (Weber *et al.*, 2023). As AI evolves, it continues to drive organizational transformation, boosting productivity and technological integration in Human Resource Management (HRM) (Black and van Esch, 2020; Köchling and Wehner, 2023; Murugesan *et al.*, 2023; Priksat *et al.*, 2023). Specifically, AI-driven tools are enhancing decision-making, improving efficiencies, and automating hiring processes (Jatobá *et al.*, 2023; Kaushal *et al.*, 2023). Recent research suggests that 24% of businesses worldwide have integrated AI into recruitment, signaling a broader trend toward embracing automated hiring processes (Meshram, 2023).

While extensive research has explored the reasons behind AI adoption in recruitment, emphasizing its benefits and strategic advantages (Allal-Chérif *et al.*, 2021; Black and van Esch, 2020; Chen, 2023; Pan *et al.*, 2023; Pillai and Sivathanu, 2020; Su *et al.*, 2021), much of this work has focused on the tools and techniques for enhancing recruitment processes, tools



such as gamified interviews (Buil *et al.*, 2020; Deterding *et al.*, 2011; Georgiou *et al.*, 2019), asynchronous video assessments (Basch *et al.*, 2022; Suen and Hung, 2023), and mobile crowdsourcing platforms (Abououf *et al.*, 2021). Despite advancements in AI tools and machine learning, the critical challenge of how to adopt AI effectively into recruitment processes remains a hindrance for many organizations. Ensuring seamless integration is essential to maximize utilization and improve recruitment outcomes (Weber *et al.*, 2023).

Recent work has begun addressing this gap by examining the role of managerial capabilities and non-technical factors in fostering AI adoption (Chowdhury *et al.*, 2023; Deepa *et al.*, 2024; Perifanis and Kitsios, 2023). For instance, Deepa *et al.* (2024), underscore the critical role of managers in driving successful AI Adoption, while Chowdhury *et al.* (2023), emphasis on developing non-technical ones such as human skills and competencies. Similarly, Chowdhury *et al.* (2024), suggest future directions for evaluating the impact of generative AI on human skills. Prior studies indicate that organizations require specific human skills to navigate AI integration (Anton *et al.*, 2020; Jöhnk *et al.*, 2021; Perifanis and Kitsios, 2023; Shet *et al.*, 2021). Supporting this view, the World Economic Forum (WEF) emphasizes that managerial readiness remains a key barrier to fully leveraging AI's potential in recruitment processes (Meho and Rogers, 2023). Yet, how to adopt AI effectively remains a significant gap, and the role of competent HR professionals is vital in navigating and driving successful AI adoption. Furthermore, strong collaboration between AI and business departments, often achieved through open innovation, is vital for successful integration (Annosi *et al.*, 2022; Jöhnk *et al.*, 2021; Murugesan *et al.*, 2023; Obradović *et al.*, 2021). Additionally, IT infrastructure (Bhatt, 2000; Shet *et al.*, 2021), and financial support (Park *et al.*, 2021), are vital for seamless AI integration. AI adoption not only transforms processes but also deeply impacts a firm's strategic direction (Allal-Chérif *et al.*, 2021).

To foster sustained performance, tangible and intangible resources of the organization need to be strategically aligned (Peteraf *et al.*, 2013; Shet and Pereira, 2021). In the context of AI adoption, RBV is essential in understanding how firms can harness these resources to effectively implement AI-driven recruitment processes. Building on RBV, the Dynamic Capabilities Framework (DCF) emphasizes the importance of a firm's ability to integrate, build, and reconfigure its resources to respond to rapidly changing environments (Graham and Moore, 2021; Teece *et al.*, 1997). The research seeks to answer the following questions:

RQ1. How do intangible resources (HR competencies and open innovation) and tangible resources (financial support and IT infrastructure) influence managerial and innovation capabilities for achieving AI adoption in recruitment processes?

RQ2. How does AI adoption in recruitment processes, supported by intangible and tangible resources, impact organizational competitive advantage?

This study demonstrates how HR competencies, open innovation, financial support, and IT infrastructure drive AI adoption, leading to improved talent acquisition and competitive advantage. By integrating RBV (Jay, 1991) and DCF (Teece *et al.*, 1997), this research provides strategic insights into managing the complexities of AI adoption and achieving long-term competitive advantage (Saura *et al.*, 2022).

In the following sections, we first review the existing literature to establish the foundation of our model in Section 2. Based on this review, we develop hypotheses to address the research questions in Section 3. Section 4 outlines the study's methodology, followed by the data analysis in Section 5. In Section 6, we discuss the findings and their implications, and finally, Section 7 presents the conclusions of the study.

2. Literature review

2.1 AI and human resource management (HRM)

AI has emerged as a transformative force in HRM, providing significant benefits to organizations (Jatobá *et al.*, 2023; Kaushal *et al.*, 2023; Weber *et al.*, 2023). One notable

definition describes AI as “the frontier of computational advancements mimicking human intelligence to address increasingly complex decision-making problems” (Berente *et al.*, 2021; Gilch and Sieweke, 2021). HRM to embrace technology and meet modern workforce demands (Horodyski, 2023a, b; Köchling and Wehner, 2023; Shet *et al.*, 2021). AI-driven tools streamline recruitment processes by screening resumes, scheduling interviews, and conducting preliminary assessments via chatbots and video platforms. These advancements improve efficiency and reduce operational costs (Allal-Chérif *et al.*, 2021; Basch *et al.*, 2022). However, ethical and legal challenges remain significant. Algorithmic bias, discrimination, and lack of explainability in AI systems can perpetuate inequalities, requiring HR to address fairness and transparency proactively (Köchling and Wehner, 2020; Lee and Cha, 2023). Compliance with data protection regulations, such as the General Data Protection Regulation (GDPR), is also essential to safeguard candidate privacy and ensure lawful data processing (Berente *et al.*, 2021; Gonzalez *et al.*, 2019; Kammerer, 2022). These challenges highlight the critical role of HR competency in refining AI tools to meet ethical standards and align with organizational objectives (Black and van Esch, 2020; Chen, 2023).

Table 1 reinforces the importance of HR competency and open innovation in successfully adopting AI in recruitment. HR competency is essential for managing AI systems, particularly in addressing fairness, transparency, and ethical compliance. For instance, HR professionals need to mitigate risks of bias and unfairness through careful oversight and regular audits (Kelan, 2023; Noble *et al.*, 2021). Additionally, trust in AI systems is crucial, as skepticisms about algorithmic recommendations remains a barrier. This can be addressed by prioritizing explainability and transparency, which help align AI tools with organizational values and build trust among recruiters (Horodyski, 2023a; Lacroux and Martin, 2022; Lee and Cha, 2023). Furthermore, the rapid evolution of AI demands that HR professionals acquire new competencies, such as technical expertise and data-driven decision-making skills, to maximize AI’s potential while preserving the human element in recruitment (Chowdhury *et al.*, 2023; Deepa *et al.*, 2024).

Open innovation complements HR competency by fostering collaboration with AI vendors and internal IT teams to customize AI tools that meet organizational needs (Rihayana *et al.*, 2023). External vendors play a critical role in bridging technological gaps, especially for organizations lacking in-house expertise to design and optimize advanced AI systems adoption (Gilch and Sieweke, 2021; Van den Broek *et al.*, 2021). This collaborative approach enables access to cutting-edge innovations while ensuring that AI tools align with ethical standards and operational goals (Black and van Esch, 2020; Chen, 2023). Combining HR competency with open innovation provides a robust foundation for addressing the complexities of AI adoption, justifying their inclusion as critical variables for this study.

2.2 Resource-based view (RBV) theory

RBV emerged in management strategy literature as a framework for evaluating a company’s market position by assessing its resource base. Initially proposed by Penrose (1959), the concept emphasized the need to consider a broader range of resources in firm analysis. Jay (1991) further developed RBV theory by introducing the idea that not all resources are equal; their value and strategic significance depend on their being Valuable, Rare, Inimitable, and Non-substitutable (VRIN). In the context of technology adoption, SMEs can leverage VRIN by prioritizing specialized, cost-effective technologies tailored to niche markets, thereby creating value through uniqueness and contextual relevance. Conversely, large firms, equipped with greater resources, fulfill VRIN criteria by investing in scalable and proprietary technologies, seamlessly integrating them across diverse operations to enhance strategic impact (Barney, 2001). Similarly, Grant (1991) categorized resources into tangible, intangible, and personnel-based types, which together build organizational capabilities.

HR competencies, as intangible resources, are key to AI adoption (Alkaraan *et al.*, 2024; King, 2016; Nankervis *et al.*, 2021). Along with human skills, organizations must select the

Table 1. Studies on AI adoption and recruitment

Sl. No	Author	Methodology	Perspective (candidate/ company)	Key theme	Findings (AI in recruitment)	Role of HR competency and collaboration role
1	Noble et al. (2021)	Experimental (2x3 design)	Candidate	Looks at AI's effect on fairness and perceptions of justice	Negative: AI reduces fairness in hiring	HR need to ensure fairness in AI-driven screening processes
2	Mirowska and Mesnet (2022)	Qualitative	Candidate	Explores applicant reactions to AI-evaluated interviews	Negative: Applicants prefer human involvement over AI	Role of HR is to balance AI with human oversight to satisfy candidate preferences
3	Lacroux and Martin (2022)	Experimental	Company	Examines recruiters' trust in AI during resume screening	Negative: Algorithmic recommendations face skepticism	HR needs to address trust and reliability issues with AI tools
4	Kelan (2023)	Theoretical Review	Company	Examines bias and fairness issues in AI-driven recruitment	Negative: Bias in AI-based hiring	HR need to mitigate biases in AI systems
5	Su et al. (2021)	Experimental Design	Company	Explores how AI predicts behavioral competencies from facial expressions in interviews	Positive: AI for predicting competencies via facial expressions	HR facilitates AI integration for competency assessments
6	Van Esch and Black (2019)	Conceptual	Company	Highlights the use of AI to enhance recruitment processes, focusing on efficiency	Positive: Promotes efficiency and effectiveness in recruitment	HR's role is emphasized in managing AI integration
7	Pillai and Sivathanu (2020)	Quantitative	Company	Discusses AI adoption for better HR efficiency in IT talent acquisition	Positive: Efficiency and cost-effectiveness in talent acquisition	HR's capability to integrate AI effectively for talent management is critical
8	Lee and Cha (2023)	Conceptual	Company	Examines how explainability can improve AI trustworthiness in recruitment	Positive: Explainability of AI systems increase trust	HR should ensure AI tools are explainable and transparent
9	Gilch and Sieweke (2021)	Qualitative	Company	Focuses on recruiting digital talent and AI's role in digital transformation	Positive: AI supports recruiting digital talent for transformation	HR needs to manage AI tools to target the right talent
10	Horodyski (2023a, b)	Quantitative	Company	Examines recruiters' views on AI tools in recruitment	Mixed: Efficiency gains but lack of human judgment	HR competency is needed for effective AI use and balancing human judgment

(continued)

Table 1. Continued

Sl. No	Author	Methodology	Perspective (candidate/company)	Key theme	Findings (AI in recruitment)	Role of HR competency and collaboration role
11	Da Motta Veiga et al. (2023)	Mixed-methods	Candidate	Explores how AI ethics impact applicant perception	Positive: Ethical AI increases attractiveness to applicants	HR should emphasize ethical AI practices in recruitment
12	Suen and Hung (2023)	Experiment	Candidate	Investigates trust factors in AI-driven video interviews	Mixed: AI can build trust if transparency is ensured	HR should ensure transparency and clear communication in AI use
13	Chen (2023)	Qualitative	Company	Explores how recruiters collaborate with AI systems to reduce bias	Mixed: AI removes biases but raises concerns about costs and control	HR must work closely with AI developers to manage costs and ensure system alignment with recruitment needs
14	Black and van Esch (2020)	Conceptual	Company	Highlights the importance of collaboration with AI vendors to optimize recruitment tools	Positive: AI is crucial for improving recruitment processes	HR professionals need to collaborate with AI vendors to ensure that systems meet organizational goals
15	Allal-Chérif et al. (2021)	Case study	Company	Discusses how organizations collaborate with AI vendors to streamline talent management processes	Positive: Increases efficiency in talent identification and retention	HR plays a role in ensuring that AI systems developed by vendors align with organizational objectives
16	Chowdhury et al. (2023)	Systematic Literature Review	Company	Proposes AI capabilities needed to succeed in HR processes	Positive (AI unlocks competitive advantage)	HR needs to develop new skills and integrate AI for operational success
17	Deepa et al. (2024)	Systematic Literature Review	Company	Explores the impact of AI technologies on HR competencies	Positive (AI enhances HRM capabilities)	HR must build new competencies to handle AI in recruitment and performance management
Source(s): Authors' own work						

right technology and developers for successful integration (Allal-Chérif *et al.*, 2021). Open innovation, especially through external partnerships (Annosi *et al.*, 2022; Bahoo *et al.*, 2023). However, AI adoption in HR comes with challenges, requiring long-term investment in hardware, software, infrastructure, and resources (Graham and Moore, 2021). As tangible resources, firms must allocate financial resources to support AI adoption and optimize operational capabilities (Park *et al.*, 2021). Additionally, successful implementation requires restructuring resources, infrastructure, particularly during periods of organizational transformation (Bhatt, 2000; Gao *et al.*, 2019).

2.3 Dynamic capability framework (DCF)

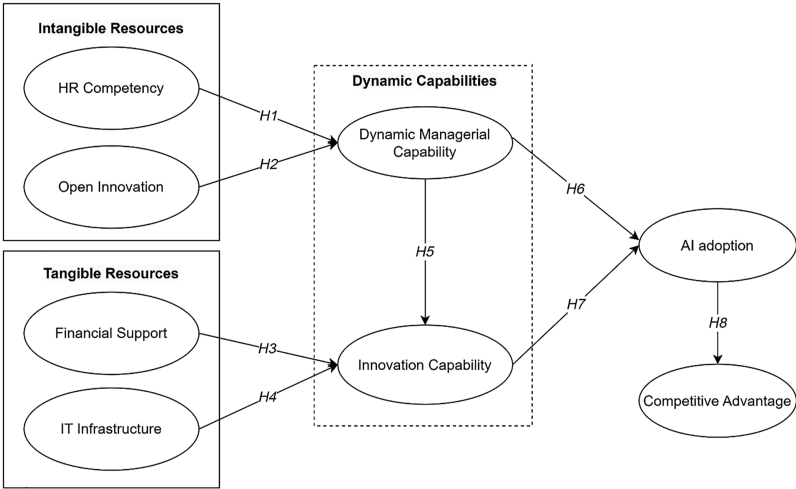
Dynamic capabilities go beyond the VRIN resources emphasized by the RBV (Jay, 1991). They focus on a firm's ability to integrate, build, and reconfigure internal and external competencies to adapt to dynamic environments, thereby ensuring competitive advantage (Teece *et al.*, 1997). They empower a firm to modify its strategy and allocate resources effectively, ensuring the maintenance and sustainability of a competitive advantage (Eisenhardt and Martin, 2000; Graham and Moore, 2021). The Dynamic Managerial Capability (DMC) theory builds on DCF by offering a framework to understand how managers influence strategic change in uncertain environments (Adner and Helfat, 2003; Helfat and Martin, 2015). DMC is defined as "the abilities with which managers construct, integrate, and reconfigure organizational resources and competencies" (Adner and Helfat, 2003).

DMC comprises three key components: Managerial Human Capital (MHC), Managerial Social Capital (MSC), and Managerial Cognition (MC) (Adner and Helfat, 2003; Helfat and Martin, 2015; Heubeck, 2023). MHC refers to managers' skills and knowledge essential for effectively implementing AI adoption. Managers with strong MHC can adopt advanced recruitment technologies, improving organizational performance and securing a competitive advantage (Peteraf *et al.*, 2013). MSC enables managers to access internal and external resources and networks, fostering the exchange of information and best practices related to digital transformation (Greven *et al.*, 2023; Heubeck, 2023; Kuzior *et al.*, 2023). Lastly, MC involves the mental models, beliefs, and cognitive processes that shape how managers perceive, interpret, and react to stimuli in their environment (Adner and Helfat, 2003). In the context of HRM, Deepa *et al.* (2024), emphasize the dynamic interplay of MHC, MSC, and MC in AI adoption. MHC provides the expertise to implement AI systems, MSC facilitates resource access and vendor collaboration for seamless integration, and MC aids strategic decision-making by aligning AI with organizational goals. Together, these capabilities enable adaptive management of complexities, ensuring efficient AI integration. Figure 1 illustrates how RBV resources are leveraged through the DCF to drive AI adoption and competitive advantage.

3. Hypotheses development

3.1 HR competency

Competency refers to the combination of skills, knowledge, and abilities that enable individuals to perform their roles effectively (Long *et al.*, 2013). In the context of HR, competencies encompass the specific expertise required by HR professionals to manage emerging technologies and foster innovation (Jatobá *et al.*, 2023; Nankervis *et al.*, 2021). As organizations increasingly evolving, developing these competencies becomes essential for HR professionals to navigate and implement AI-driven processes (Anton *et al.*, 2020). Lack of capabilities hinders the organization growth (Gao *et al.*, 2019). However, continuous reskilling and upskilling of HR professionals enhances MHC, a core component of DMC (Helfat and Martin, 2015; Heubeck, 2023). By strengthening MHC, HR competencies positively influence DMC, enabling HR teams to align AI systems with organizational goals and drive innovation



Source(s): Authors' own work

Figure 1. Proposed model for contribution of AI in recruitment

and operational efficiency (Anton *et al.*, 2020; Basch *et al.*, 2022; Ratnam and Devi, 2024; Shet *et al.*, 2021). Based on this understanding, we hypothesize:

H1. HR competency has a positive influence on Dynamic Managerial Capabilities, facilitating AI adoption in recruitment processes.

3.2 Open innovation

Open innovation plays a critical role in fostering collaborative idea generation and integrating internal and external knowledge, which is pivotal for organizations adopting AI technologies. Recent studies underscore its strategic significance in navigating the rapidly evolving technological landscape. For instance, Jia *et al.* (2023) emphasize that open innovation expands knowledge sources for developing new products, services, or processes, enabling organizations to transcend traditional R&D boundaries. HR professionals, as key enablers, collaborate with external partners to introduce innovative practices. For instance, collaboration between HR professionals and AI developers fosters mutual learning, combining domain expertise with technical knowledge to enable effective AI adoption (Gilch and Sieweke, 2021; Van den Broek *et al.*, 2021). Similarly, interdepartmental collaboration between HR, IT, and other key departments ensures the seamless integration of AI solutions, aligning them with organizational goals and achieving impactful outcomes (Chowdhury *et al.*, 2023). Firms like P&G and NASA effectively leveraged external collaborations for technological advancements and adaptable business models (Obradović *et al.*, 2021).

As managerial roles adapt to automation, cross-functional collaboration becomes even more vital to align innovation practices with broader organizational strategies (Langer *et al.*, 2021; Rihayana *et al.*, 2023). MSC, a key component of DMC, supports open innovation by leveraging networks to access critical external knowledge and technologies. This enhances AI adoption and drives competitive advantage (Greven *et al.*, 2023; Jia *et al.*, 2023). These elements illustrate how open innovation and collaboration serve as catalysts for leveraging AI technologies to enhance organizational performance. Based on these insights, we propose the following hypothesis.

H2. Open innovation positively influences Dynamic Managerial Capabilities, supporting AI adoption in recruitment.

3.3 Financial support

Financial support refers to the liquid funding that firms allocate to drive innovation and implement technological advancements (Voss *et al.*, 2008). A survey highlighted that 68% of HR professionals identified lack of financial resources as a primary barrier to adopting AI in hiring processes (Park *et al.*, 2021). Adequate financial backing enhances a firm's ability to invest in and sustain innovation. Previous studies, particularly in sectors like the automobile industry, have shown that financial resources positively impact Innovation Capability by enabling firms to invest in critical technologies and drive growth (Voss *et al.*, 2008; Wang *et al.*, 2016). In the context of AI adoption, financial support is crucial for making investments in advanced AI systems, infrastructure, and upskilling HR professionals to facilitate seamless integration.

H3. Financial support has a positive influence on innovation capabilities, particularly in facilitating AI adoption.

3.4 IT infrastructure

IT infrastructure forms the backbone for integrating hardware, software, and data into a unified system that supports seamless information flow across an organization (Muazu and Abdulmalik, 2021). It is essential for the successful deployment and management of AI technologies, enabling efficient distribution of data and media to stakeholders (Park *et al.*, 2021; Shet *et al.*, 2021). A robust IT infrastructure positively influences a firm's innovative capability and competitive advantage (Rahman *et al.*, 2022). A well-developed IT infrastructure supports organizations in their ability to adopt and implement new technologies by providing the technological foundation required for AI-driven processes (Bharadwaj *et al.*, 1999; Bhatt, 2000; Muazu and Abdulmalik, 2021). Variations in IT infrastructure can significantly impact how effectively new technologies are adopted, influencing an organization's ability to leverage AI for competitive advantage (Wang *et al.*, 2016). Based on these understanding, we propose the following hypothesis:

H4. IT infrastructure positively influences innovation capability and the successful integration of AI technologies.

3.5 Dynamic managerial capability (DMC)

DMC are instrumental in driving the successful adoption and implementation of AI technologies, particularly in recruitment processes (Deepa *et al.*, 2024). DMC refers to "managers' abilities to integrate, build, and reconfigure internal and external competencies to rapidly address changes in the environment, especially those related to the adoption and application of AI technologies (Helfat and Martin, 2015; Heubeck, 2023). These capabilities play a critical role in determining how effectively organizations can adopt and leverage AI for strategic advantage (Helfat and Martin, 2015; Majhi *et al.*, 2023), enabling HR leaders to manage the complexities of AI integration. For example, Böhmer and Schinnenburg (2023) emphasize that robust managerial capabilities are essential in AI-powered HR management, as they support the development of organizational competencies critical for AI adoption. Similarly, Alkaraan *et al.* (2024) highlight that DMC are pivotal for fostering innovation and technology adoption in organizations.

Earlier studies indicates that a firm's sensing dynamic capability is crucial for initiating, exploring, and executing management innovations (Liao *et al.*, 2007). Integrative dynamic capabilities, on the other hand, play a key role in the early stages of innovation adoption, such as establishing proposals and facilitating organizational adoption processes (H. Chen *et al.*, 2021). These insights underline the importance of DMC particularly the ability to sense opportunities, integrate knowledge, and reconfigure resources for driving innovation within firms. Based on these findings, we propose the following hypotheses.

- H5. Dynamic Managerial Capability positively influences a firm’s innovation capability.
- H6. Dynamic Managerial Capability positively influences the successful adoption and utilization of AI in recruitment processes.

3.6 Innovation capability

Innovation capability refers to a firm’s ability to develop new technologies or significantly improve existing ones (Cheng and Huizingh, 2014; Wonglimpiyarat, 2010), enabling organizations to quickly adapt to market changes through product and process innovations. Developing IC is crucial for organizational growth and success (Cheng and Huizingh, 2014; Liao et al., 2012; Su et al., 2021). IC involves generating or adopting innovations, supported by internal capabilities (Saunila and Ukko, 2014), and enables continuous transformation and value creation (Kelan, 2023; Saunila and Ukko, 2014). According to Mendoza-Silva (2020), IC encompasses multiple organizational dimensions that foster ongoing innovation. In today’s fast-paced technological environment, strong IC is essential for integrating and leveraging AI technologies (Bahoo et al., 2023). Firms with robust IC are better positioned to adopt AI in talent acquisition, improving efficiency and gaining a competitive advantage (Bansal et al., 2023; Cheng and Huizingh, 2014). Based on these insights, we propose the following hypothesis:

- H7. Innovation capability is positively related to the adoption of AI in recruitment.

3.7 AI adoption to competitive advantage

The integration of AI in recruitment processes not only boosts an organization’s technological reputation but also enhances operational efficiency through automation and cost reduction (Saura et al., 2022). AI-driven tools simplify candidate assessments, enabling HR managers to process larger talent pools more effectively, promote diversity, and select candidates based on standardized, bias-reducing criteria (Pan et al., 2023). By facilitating data-driven decision-making, AI empowers HR professionals to make more informed hiring choices, resulting in a more skilled and qualified workforce. Krakowski et al. (2023) investigated the impact of AI on human capabilities using chess players as a model. Their study demonstrated how AI augments human abilities, providing organizations with a competitive edge in talent acquisition and management. Given these insights, we propose the following hypothesis:

- H8. The adoption of AI positively influences an organization’s competitive advantage.

4. Research methodology

4.1 Data collection

This study utilized a cross-sectional design to explore AI adoption among HR professionals, offering a snapshot of current practices and attitudes at a single point in time. This design facilitated efficient data collection and enabled the identification of key trends and relationships without the need for extended timeframes. Data collection focused on South India and targeted HR professionals actively engaged in Talent Acquisition. Purposive sampling was employed to recruit HR professionals with direct experience in emerging technologies, ensuring the validity of findings by focusing on knowledgeable respondents. Participants represented a range of industries, enabling a robust quantitative analysis that highlights sector-specific dynamics in AI adoption. The survey was distributed individually through a combination of personal interactions and recruitment association networks, ensuring engagement and accurate responses. We collected demographic data, including Gender, Industry, Position in the Organization, Age, and Qualification, at the outset of the questionnaire. Demographic details of respondents are given in Table 2.

Table 2. Demographic details of respondents

Characteristics	No. (N = 290)	%
<i>Gender</i>		
Male	119	41%
Female	171	59%
<i>Industry</i>		
IT-software/software services	124	43%
Manufacturing/FMCG	91	31%
Medical/healthcare/hospital	41	14%
Banking/financial services	34	12%
<i>Grade/position in the organization</i>		
Manager level	124	43%
Executive level	166	57%
<i>Age</i>		
21–27 (Gen Z)	71	24%
28–43 (Millennials)	219	76%
<i>Qualification</i>		
Graduation	32	11%
Postgraduation	258	89%

Source(s): Authors' own work

At the outset, respondents were asked to provide their informed consent before completing the survey questionnaire. The researchers ensured that all participants were fully informed about the research objectives and the procedures for storing and protecting the collected data throughout the study. Measures were also taken to guarantee the confidentiality of all participants. A total of 886 questionnaires were distributed to HR individuals, resulting in 354 responses, representing a response rate of 40%. After excluding 64 incomplete or invalid responses, 290 valid questionnaires remained, yielding a final effective response rate of 32% for inclusion in statistical analysis. Responses were captured on a 5-point Likert scale, ranging from 1 (Strongly Disagree) to 5 (Strongly Agree).

The study adopted constructs from established and reliable scales to measure the key variables. The specific measurement items for each construct are detailed in the questionnaire in [Appendix 1](#). To ensure the robustness of the questionnaire, validity and reliability were confirmed through multiple assessments. Content validity was established via expert reviews from 8 HR managers and 5 academic experts. A pilot study further refined the questionnaire based on initial feedback.

4.1.1 Data analysis. Data were analyzed using Partial Least Squares Structural Equation Modeling (PLS-SEM 4), chosen for its predictive accuracy and ability to handle complex models with small to medium sample sizes. PLS-SEM allows simultaneous analysis of measurement and structural models, making it ideal for exploratory research. The measurement model was assessed to confirm the reliability and validity of the constructs through Cronbach's Alpha, Composite Reliability (CR), and Average Variance Extracted (AVE). Factor loadings for all measurement items were assessed, with most exceeding the threshold of 0.7, indicating strong indicator reliability. However, OI1 (0.695) and FS3 (0.678) slightly fell below this threshold but were retained due to their theoretical importance and acceptable proximity to the cutoff. AVE values were above 0.5, confirming convergent validity. Discriminant validity was further evaluated using the Fornell-Larcker criterion, ensuring that the square root of the AVE for each construct was greater than its correlations with other constructs. Additionally, the HTMT ratio values were below 0.85, providing further evidence of discriminant validity. Model fit was evaluated using the Standardized Root Mean

Square Residual (SRMR), which was below the threshold of 0.08, demonstrating an acceptable fit for the data.

5. Results

5.1 Common method bias (CMB)

Common method bias (CMB) was evaluated using Harman’s single-factor test. The results indicated that a single factor accounted for 31.21% of the total variance, which is well below the 50% threshold, confirming that CMB was not a significant concern in this study (Fuller et al., 2016).

5.2 Measurement model

The measurement model was evaluated using PLS-SEM 4 by assessing outer loadings, composite reliability (CR), average variance extracted (AVE), and discriminant validity (Hair et al., 2019). Cronbach’s Alpha and CR scores for all constructs exceeded the 0.7 threshold, confirming strong internal consistency (Hair et al., 2019). Convergent validity was confirmed with AVE values greater than 0.5 (Hair et al., 2017) (see Table 3).

Discriminant validity was assessed using the Fornell-Larcker criterion (see Table 4) and Heterotrait-Monotrait Ratio (HTMT) (see Table 5). The square root of the AVE for each construct exceeded its correlation with any other construct, and all HTMT ratios were significantly below the 0.85 threshold (Henseler et al., 2015). Multicollinearity was examined using Variance Inflation Factors (VIFs), with all values below 3.3, indicating no significant multicollinearity concerns.

Table 3. Validity and reliability

Constructs	Items	Factor loadings	Cronbach alpha	CR	AVE	VIF
HR competency (HR Comp)	HRCOM1	0.857	0.816	0.829	0.730	1.734
	HRCOM2	0.894				2.422
	HRCOM3	0.811				1.815
Open innovation (OI)	OI1	0.695	0.822	0.864	0.575	1.766
	OI2	0.761				1.795
	OI3	0.722				1.683
	OI4	0.845				2.918
	OI5	0.759				1.302
Managerial Capability (MC)	MC1	0.798	0.690	0.697	0.619	1.488
	MC2	0.838				1.526
	MC3	0.720				1.206
Financial support (FS)	FS1	0.856	0.760	0.834	0.665	1.532
	FS2	0.895				1.845
	FS3	0.678				1.468
IT infrastructure (ITI)	ITI1	0.746	0.741	0.746	0.661	1.302
	ITI2	0.841				1.642
	ITI3	0.848				1.715
Innovation capability (IC)	IC1	0.813	0.793	0.800	0.707	1.649
	IC2	0.870				1.834
	IC3	0.839				1.608
AI adoption (AIA)	AIA1	0.754	0.756	0.803	0.669	1.396
	AIA2	0.830				1.634
	AIA3	0.870				1.628
Competitive Advantage (CA)	CA1	0.870	0.812	0.839	0.724	1.656
	CA2	0.853				2.014
	CA3	0.829				1.803

Source(s): Authors’ own work

Table 4. Fornell-Larcker criterion

Constructs	AIA	FS	HRCOM	IC	ITI	MC	CA	OI
AI adoption	0.818							
Financial support	0.113	0.815						
HR competency	0.213	0.203	0.855					
Innovation capability	0.348	0.282	0.271	0.841				
IT infrastructure	0.101	0.347	0.231	0.409	0.813			
Managerial capability	0.239	0.205	0.378	0.261	0.195	0.787		
Competitive advantage	0.466	0.055	0.26	0.242	0.159	0.295	0.851	
Open innovation	0.289	0.343	0.447	0.29	0.419	0.25	0.758	0.758

Source(s): Authors' own work**Table 5.** Discriminant validity (HTMT)

Constructs	AIAI	FS	HRCOM	IC	ITI	MC	MOD	OI
AI adoption	0.000							
Financial support	0.157							
HR competency	0.259	0.247						
Innovation capability	0.439	0.327	0.330					
IT infrastructure	0.125	0.475	0.300	0.532				
Managerial capability	0.321	0.258	0.500	0.348	0.271			
Competitive advantage	0.559	0.093	0.321	0.296	0.202	0.389		
Open innovation	0.330	0.344	0.403	0.553	0.376	0.507	0.298	0.000

Source(s): Authors' own work

5.3 Structural model results

Following the confirmation of the measurement model's validity and reliability, the structural model was analyzed to evaluate the hypotheses. Collinearity diagnostics indicated that tolerance levels for predictor constructs were comfortably below the VIF threshold of 5, as recommended by [Hair et al. \(2019\)](#). The path coefficients confirmed that HR competency and open innovation positively influenced managerial capability. Financial support and IT infrastructure significantly impacted innovation capability. Managerial capability affected both innovation capability and AI adoption, while innovation capability also influenced AI adoption. Finally, AI adoption significantly enhanced competitive advantage. These results are summarized in [Table 6](#).

Table 6. Path analysis

Relationships	β	<i>t</i> -statistics	<i>p</i> -values	Results
AIAI $\rightarrow$ CA	0.466	8.709	0.000	Supported
FS $\rightarrow$ IC	0.132	1.964	0.050	Supported
HRCOM $\rightarrow$ MC	0.266	5.108	0.000	Supported
IC $\rightarrow$ AIAI	0.306	4.607	0.000	Supported
ITI $\rightarrow$ IC	0.330	5.206	0.000	Supported
MC $\rightarrow$ AIAI	0.159	2.734	0.006	Supported
MC $\rightarrow$ IC	0.170	2.796	0.005	Supported
OI $\rightarrow$ MC	0.327	6.300	0.000	Supported

Note(s): * $p < 0.05$ – Significant**Source(s):** Authors' own work

6. Discussion

The findings confirm that both tangible and intangible resources significantly contribute to AI adoption and bolster organizational competitive advantage, aligning with the principles of RBV and DCF. [H1](#) validates a positive relationship between HR competency and DMC ([Anton et al., 2020](#); [Chowdhury et al., 2023](#); [Deepa et al., 2024](#); [Perifanis and Kitsios, 2023](#); [Shet et al., 2021](#)). Similarly, [H2](#) highlights the pivotal role of open innovation in enhancing DMC ([Annosi et al., 2022](#); [Chesbrough, 2006](#); [Greven et al., 2023](#); [Jöhnk et al., 2021](#); [Murugesan et al., 2023](#)). Financial support and IT infrastructure, as shown in [H3](#) and [H4](#), play crucial roles in fostering innovation capabilities ([Shet et al., 2021](#); [Voss et al., 2008](#)). [H5](#) and [H6](#) highlight that DMC enhances innovation capabilities and drives AI adoption ([Alkaraan et al., 2024](#); [Helfat and Martin, 2015](#)). [H7](#) confirms that innovation capabilities positively impact AI adoption ([Wonglimpiyarat, 2010](#)), and [H8](#) confirms that AI adoption positively impacts competitive advantage ([Saura et al., 2022](#)). Model details are in [Appendix 2](#).

In addressing [RQ1](#), the study shows that both intangible resources (HR competencies, open innovation) and tangible resources (financial support, IT infrastructure) play crucial roles in enhancing managerial and innovation capabilities for AI adoption in recruitment processes. HR competencies and open innovation significantly enhance managerial capabilities, while financial support and IT infrastructure provide the backbone for fostering innovation. Addressing [RQ2](#), the study illustrates that AI adoption in recruitment processes, driven by both intangible and tangible resources, significantly strengthens organizational competitive advantage. The findings underscore the importance of continuous evolution and adaptability, key tenets of the DCV, as sources of sustained strategic advantage. The DCV highlights that continuous evolution and adaptability can be sources of sustained competitive advantage. Our results support by showing that firms that actively develop and renew their resources, particularly through AI adoption, are better positioned to stay competitive in rapidly changing markets.

6.1 Theoretical implications

This study offers several important theoretical contributions. First, it demonstrates that both tangible and intangible organizational resources are critical for effectively managing AI adoption, particularly in enhancing talent acquisition and decision-making processes. By extending the RBV, this research underscores the significance of integrating resources in technological contexts, highlighting how resource orchestration can influence successful AI implementation. Second, the study emphasizes the essential role of DMC in driving innovation and adapting to rapid regulatory and technological changes. It contributes to the literature by illustrating how managerial competencies, including sensing opportunities, aligning with legal frameworks, and reconfiguring organizational resources, are pivotal for innovation and competitiveness. This expands the application of DMCs in the context of emerging technologies, offering actionable insights into their evolution and practical strategies for enhancement. Third, the research highlights the critical role of internal resources, particularly HR competencies, in organizational transformation. It demonstrates that organizations must proactively develop internal capabilities, such as upskilling HR professionals and fostering collaboration, rather than relying solely on external forces.

By integrating RBV and DMC frameworks, this study provides a comprehensive understanding of how organizations can navigate the complexities of AI-driven recruitment. It offers a novel perspective on resource reconfiguration and dynamic capabilities in response to technological change. Furthermore, it contributes to the literature on human-machine collaboration by illustrating how AI complements existing HR resources, creating a hybrid model where technology and human expertise enhance organizational performance in tandem. Finally, this study presents a new lens on sustaining competitive advantage in dynamic

environments. By showcasing the integration of AI-driven strategies with human resources, it broadens the scope of RBV and DMC in the digital age, providing a framework for organizations to align their technological and human capital to drive sustained success.

6.2 Managerial implications

This study provides critical insights for HR professionals on how to build competencies required for AI adoption effectively, addressing two key objectives. First, HR competencies, combined with open innovation, enable professionals to effectively leverage AI recruitment technologies, enhancing speed, accuracy, and access to a broader talent pool, fostering seamless AI integration with external partners. Second, successful adoption of AI technologies gives organizations a competitive edge by enhancing operational efficiency, particularly in environments where open resources facilitate rapid technology adoption. Companies like P&G using external ideas to enhance product strategies and NASA leveraging crowdsourcing to drive innovation demonstrate the importance of aligning collaborative efforts with strategic goals (Obradović *et al.*, 2021).

To implement these recommendations effectively, HR teams need to address not only technical and cultural barriers but also significant ethical and legal challenges. For instance, algorithmic bias, which can perpetuate discrimination, must be mitigated through regular audits human interaction and ethical algorithm design to ensure fairness and inclusivity. From a legal perspective, compliance with data protection laws such as GDPR is critical. Organizations must ensure lawful data processing, obtain explicit consent, and safeguard candidate privacy to align with regulatory standards (Hauer *et al.*, 2021; Kammerer, 2022).

To address these challenges, organizations should foster a culture that embraces AI by providing change management training and encouraging collaboration between HR and IT teams. Upskilling HR professionals with technical and analytical skills is crucial to bridging knowledge gaps in managing AI systems effectively. Competent HR professionals play a pivotal role in mitigating legal and ethical issues through augmented recruitment processes by ensuring compliance with regulations, fostering transparency, and maintaining fairness in AI-driven systems. Managerial competencies must extend beyond technical expertise to include leadership in integrating AI into workflows and decision-making processes. Furthermore, AI adoption enhances HR capabilities and contributes to the organization's intellectual capital, solidifying its competitive position.

6.3 Limitations and future research directions

This study acknowledges several limitations that provide valuable opportunities for future research. One significant limitation is the lack of transparency in AI-based systems, which can undermine managerial trust in AI outputs. Addressing this issue requires further exploration of strategies to enhance explainability and trustworthiness in AI applications. Another limitation lies in the cross-sectional design of the study, which captures data at a single point in time and does not account for the evolving nature of AI adoption. Future longitudinal studies could provide deeper insights into the long-term effects of AI adoption across various industries and regions. Such studies would illuminate how organizations adapt to technological advancements and regulatory changes over time, offering more dynamic perspectives on AI integration. Dynamic capabilities within the frameworks of RBV, dynamic capability theory, and adoption intention also warrant further investigation. Future studies could explore constructs such as resistance to cultural change, lack of technical skills in HR professionals, implementation costs, business transformation, and multi-stakeholder collaboration. Understanding how these factors influence AI adoption in diverse organizational contexts could yield actionable insights for both researchers and practitioners. Finally, the study's focus on a developed economy and the creative industries may limit the generalizability of its findings.

7. Conclusion

This study underscores the critical role of both tangible and intangible resources in facilitating AI adoption within recruitment processes and highlights their impact on gaining a competitive advantage. By integrating the RBV and DCF, the research demonstrates that HR competencies, open innovation, financial support, and IT infrastructure are key drivers of AI implementation success. These findings suggest that to fully leverage the potential of AI, organizations must strategically align their human and technological resources. HR professionals play a pivotal role not only in managing AI-driven recruitment tools but also in ensuring fairness, transparency, and avoiding bias. They work closely with AI developers to configure the most suitable AI tools for recruitment, ensuring that these AI tools meet the specific needs of the organization while upholding ethical standards.

References

- Abououf, M., Otrók, H., Mizouni, R., Singh, S. and Damiani, E. (2021), "How artificial intelligence and mobile crowd sourcing are inextricably intertwined", *IEEE Network*, Vol. 35 No. 3, pp. 252-258, doi: [10.1109/MNET.011.2000516](https://doi.org/10.1109/MNET.011.2000516).
- Adner, R. and Helfat, C.E. (2003), "Corporate effects and dynamic managerial capabilities", *Strategic Management Journal*, Vol. 24, 10 SPEC ISS, pp. 1011-1025, doi: [10.1002/smj.331](https://doi.org/10.1002/smj.331).
- Agarwal, A. (2023), "AI adoption by human resource management: a study of its antecedents and impact on HR system effectiveness", *Foresight*, Vol. 25 No. 1, pp. 67-81, doi: [10.1108/FS-10-2021-0199](https://doi.org/10.1108/FS-10-2021-0199).
- Alkaraan, F., Elmarzouky, M., Hussainey, K., Venkatesh, V.G., Shi, Y. and Gulko, N. (2024), "Reinforcing green business strategies with Industry 4.0 and governance towards sustainability: natural-resource-based view and dynamic capability", *Business Strategy and the Environment*, Vol. 2023 No. 4, pp. 1-19, doi: [10.1002/bse.3665](https://doi.org/10.1002/bse.3665).
- Allal-Chérif, O., Yela Aránega, A. and Castaño Sánchez, R. (2021), "Intelligent recruitment: how to identify, select, and retain talents from around the world using artificial intelligence", *Technological Forecasting and Social Change*, Vol. 169, January, 120822, doi: [10.1016/j.techfore.2021.120822](https://doi.org/10.1016/j.techfore.2021.120822).
- Annosi, M.C., Martini, A., Marzi, G., Vignoli, M. and Parra, H. (2022), "How to organize for open innovation from the ground up: a microfoundations approach in a foodservice firm", *British Food Journal*, Vol. 124 No. 13, pp. 391-408, doi: [10.1108/BFJ-06-2021-0641](https://doi.org/10.1108/BFJ-06-2021-0641).
- Anton, E., Behne, A. and Teuteberg, F. (2020), "The humans behind artificial intelligence – an operationalisation of AI competencies", *Twenty-Eighth European Conference on Information Systems (ECIS)*.
- Bahoo, S., Cucculelli, M. and Qamar, D. (2023), "Artificial intelligence and corporate innovation: a review and research agenda", *Technological Forecasting and Social Change*, Vol. 188, pp. 1-79, doi: [10.1016/j.techfore.2022.122264](https://doi.org/10.1016/j.techfore.2022.122264).
- Bansal, A., Panchal, T., Jabeen, F., Mangla, S.K. and Singh, G. (2023), "A study of human resource digital transformation (HRDT): a phenomenon of innovation capability led by digital and individual factors", *Journal of Business Research*, Vol. 157, December 2022, 113611, doi: [10.1016/j.jbusres.2022.113611](https://doi.org/10.1016/j.jbusres.2022.113611).
- Barney, J.B. (2001), "Is the resource-based 'view' a useful perspective for strategic management research? Yes", *Academy of Management Review*, Vol. 26 No. 1, pp. 41-56, doi: [10.2307/259393](https://doi.org/10.2307/259393).
- Basch, J.M., Melchers, K.G. and Büttner, J.C. (2022), "Preselection in the digital age: a comparison of perceptions of asynchronous video interviews with online tests and online application documents in a simulation context", *International Journal of Selection and Assessment*, Vol. 30 No. 4, pp. 639-652, doi: [10.1111/ijsa.12403](https://doi.org/10.1111/ijsa.12403).
- Berente, N., Gu, B., Recker, J. and Santhanam, R. (2021), "Managing artificial intelligence", *Journal of Wildlife Management*, Vol. 87 No. 8, pp. 1433-1450, doi: [10.1002/jwmg.22492](https://doi.org/10.1002/jwmg.22492).

- Bharadwaj, A.S., Bharadwaj, S.G. and Konsynski, B.R. (1999), "Information technology effects on firm performance as measured by Tobin's q", *Management Science*, Vol. 45 No. 7, pp. 1008-1024, doi: [10.1287/mnsc.45.7.1008](https://doi.org/10.1287/mnsc.45.7.1008).
- Bhatt, G.D. (2000), "Exploring the relationship between information technology, infrastructure and business process re-engineering", *Business Process Management Journal*, Vol. 6 No. 2, pp. 139-163, doi: [10.1108/14637150010324085](https://doi.org/10.1108/14637150010324085).
- Black, J.S. and van Esch, P. (2020), "AI-enabled recruiting: what is it and how should a manager use it?", *Business Horizons*, Vol. 63 No. 2, pp. 215-226, doi: [10.1016/j.bushor.2019.12.001](https://doi.org/10.1016/j.bushor.2019.12.001).
- Böhmer, N. and Schinnenburg, H. (2023), "Critical exploration of AI-driven HRM to build up organizational capabilities", *Employee Relations*, Vol. 45 No. 5, pp. 1057-1082, doi: [10.1108/ER-04-2022-0202](https://doi.org/10.1108/ER-04-2022-0202).
- Buil, I., Catalán, S. and Martínez, E. (2020), "Understanding applicants' reactions to gamified recruitment", *Journal of Business Research*, Vol. 110, pp. 41-50, doi: [10.1016/j.jbusres.2019.12.041](https://doi.org/10.1016/j.jbusres.2019.12.041).
- Chen, Z. (2023), "Collaboration among recruiters and artificial intelligence: removing human prejudices in employment", *Cognition, Technology and Work*, Vol. 25 No. 1, pp. 135-149, doi: [10.1007/s10111-022-00716-0](https://doi.org/10.1007/s10111-022-00716-0).
- Chen, H., Li, L. and Chen, Y. (2021), "Explore success factors that impact artificial intelligence adoption on telecom industry in China", *Journal of Management Analytics*, Vol. 8 No. 1, pp. 36-68, doi: [10.1080/23270012.2020.1852895](https://doi.org/10.1080/23270012.2020.1852895).
- Cheng, C.C.J. and Huizingh, E.K.R.E. (2014), "When is open innovation beneficial? The role of strategic orientation", *Journal of Product Innovation Management*, Vol. 31 No. 6, pp. 1235-1253, doi: [10.1111/jpim.12148](https://doi.org/10.1111/jpim.12148).
- Chesbrough, H. (2006), "Why companies should have open business models", *MIT Sloan Management Review*, Vol. 48 No. 2, pp. 22-28.
- Chowdhury, S., Dey, P., Joel-Edgar, S., Bhattacharya, S., Rodriguez-Espindola, O., Abadie, A. and Truong, L. (2023), "Unlocking the value of artificial intelligence in human resource management through AI capability framework", *Human Resource Management Review*, Vol. 33 No. 1, 100899, doi: [10.1016/j.hrmr.2022.100899](https://doi.org/10.1016/j.hrmr.2022.100899).
- Chowdhury, S., Budhwar, P. and Wood, G. (2024), "Generative artificial intelligence in business: towards a strategic human resource management framework", *British Journal of Management*, Vol. 35 No. 4, pp. 1680-1691, doi: [10.1111/1467-8551.12824](https://doi.org/10.1111/1467-8551.12824).
- Da Motta Veiga, S.P., Figueroa-Armijos, M. and Clark, B.B. (2023), "Seeming ethical makes you attractive: unraveling how ethical perceptions of AI in hiring impacts organizational innovativeness and attractiveness", *Journal of Business Ethics*, Vol. 186 No. 1, pp. 199-216, doi: [10.1007/s10551-023-05380-6](https://doi.org/10.1007/s10551-023-05380-6).
- Deci, E.L., Vallerand, R.J., Pelletier, L.G. and Ryan, R.M. (1991), "Motivation and education: the self-determination perspective", *Educational Psychologist*, Vol. 26 Nos 3-4, pp. 325-346, doi: [10.1080/00461520.1991.9653137](https://doi.org/10.1080/00461520.1991.9653137).
- Deepa, R., Sekar, S., Malik, A., Kumar, J. and Attri, R. (2024), "Impact of AI-focussed technologies on social and technical competencies for HR managers – a systematic review and research agenda", *Technological Forecasting and Social Change*, Vol. 202, May 2023, 123301, doi: [10.1016/j.techfore.2024.123301](https://doi.org/10.1016/j.techfore.2024.123301).
- Deterding, S., O'Hara, K., Sicart, M., Dixon, D. and Nacke, L. (2011), "Gamification: using game design elements in non-gaming contexts", *Conference on Human Factors in Computing Systems - Proceedings*, pp. 2425-2428, doi: [10.1145/1979742.1979575](https://doi.org/10.1145/1979742.1979575).
- Dwivedi, Y.K., Balakrishnan, J., Das, R. and Dutot, V. (2023), "Resistance to innovation: A dynamic capability model based enquiry into retailers' resistance to blockchain adaptation", *Journal of Business Research*, Vol. 157, p. 113632, doi: [10.1016/j.jbusres.2022.113632](https://doi.org/10.1016/j.jbusres.2022.113632).
- Eisenhardt and Martin, J.A. (2000), "Dynamic capabilities: what are they?", *Strategic Management Journal*, Vol. 21, July, pp. 1-23, doi: [10.1002/1097-0266\(200010/11\)21:10/11<1105::AID-SMJ133>3.0.CO;2-E](https://doi.org/10.1002/1097-0266(200010/11)21:10/11<1105::AID-SMJ133>3.0.CO;2-E).

- Ferreira, J., Cardim, S. and Coelho, A. (2021), "Dynamic capabilities and mediating effects of innovation on the competitive advantage and firm's performance: the moderating role of organizational learning capability", *Journal of the Knowledge Economy*, Vol. 12 No. 2, pp. 620-644, doi: [10.1007/s13132-020-00655-z](https://doi.org/10.1007/s13132-020-00655-z).
- Fuller, C.M., Simmering, M.J., Atinc, G., Atinc, Y. and Babin, B.J. (2016), "Common methods variance detection in business research", *Journal of Business Research*, Vol. 69 No. 8, pp. 3192-3198, doi: [10.1016/j.jbusres.2015.12.008](https://doi.org/10.1016/j.jbusres.2015.12.008).
- Gao, S., Hakanen, E., Töytäri, P. and Rajala, R. (2019), "Digital transformation in asset-intensive businesses: lessons learned from the metals and mining industry", *Proceedings of the Annual Hawaii International Conference on System Sciences, 2019-Janua*, pp. 4927-4936, doi: [10.24251/hicss.2019.593](https://doi.org/10.24251/hicss.2019.593).
- Georgiou, K., Gouras, A. and Nikolaou, I. (2019), "Gamification in employee selection: the development of a gamified assessment", *International Journal of Selection and Assessment*, Vol. 27 No. 2, pp. 91-103, doi: [10.1111/ijsa.12240](https://doi.org/10.1111/ijsa.12240).
- Gilch, P.M. and Sieweke, J. (2021), "Recruiting digital talent: the strategic role of recruitment in organisations' digital transformation", *German Journal of Human Resource Management*, Vol. 35 No. 1, pp. 53-82, doi: [10.1177/2397002220952734](https://doi.org/10.1177/2397002220952734).
- Gonzalez, M., Capman, J., Oswald, F., Theys, E. and Tomczak, D. (2019), "'Where's the I-O?' Artificial intelligence and machine learning in talent management systems", *Personnel Assessment and Decisions*, Vol. 5 No. 3, doi: [10.25035/pad.2019.03.005](https://doi.org/10.25035/pad.2019.03.005).
- Graham, K.W. and Moore, R.S. (2021), "The role of dynamic capabilities in firm-level technology adoption processes: a qualitative investigation", *Journal of Innovation Management*, Vol. 9 No. 1, pp. 25-50, doi: [10.24840/2183-0606_009.001_0004](https://doi.org/10.24840/2183-0606_009.001_0004).
- Grant, R.M. (1991), "The resource-based theory of competitive advantage: implications for strategy formulation", *California Management Review*, Vol. 33 No. 3, pp. 114-135, doi: [10.2307/41166664](https://doi.org/10.2307/41166664).
- Greven, A., Kruse, S., Vos, A., Strese, S. and Brettel, M. (2023), "Achieving product ambidexterity in new product development: the role of middle managers' dynamic managerial capabilities", *Journal of Management Studies*, Vol. 60 No. 7, pp. 1786-1818, doi: [10.1111/joms.12886](https://doi.org/10.1111/joms.12886).
- Hair, J.F., Hult, G.T.M., Ringle, C.M., Sarstedt, M. and Thiele, K.O. (2017), "Mirror, mirror on the wall: a comparative evaluation of composite-based structural equation modeling methods", *Journal of the Academy of Marketing Science*, Vol. 45 No. 5, pp. 616-632, doi: [10.1007/s11747-017-0517-x](https://doi.org/10.1007/s11747-017-0517-x).
- Hair, J.F., Risher, J.J., Sarstedt, M. and Ringle, C.M. (2019), "When to use and how to report the results of PLS-SEM", *European Business Review*, Vol. 31 No. 1, pp. 2-24, doi: [10.1108/EBR-11-2018-0203](https://doi.org/10.1108/EBR-11-2018-0203).
- Hauer, M.P., Kevekordes, J. and Haeri, M.A. (2021), "Legal perspective on possible fairness measures – a legal discussion using the example of hiring decisions", *Computer Law and Security Report*, Vol. 42, 105583, doi: [10.1016/j.clsr.2021.105583](https://doi.org/10.1016/j.clsr.2021.105583).
- Helfat, C.E. and Martin, J.A. (2015), "Dynamic managerial capabilities: review and assessment of managerial impact on strategic change", *Journal of Management*, Vol. 41 No. 5, pp. 1281-1312, doi: [10.1177/0149206314561301](https://doi.org/10.1177/0149206314561301).
- Henseler, J., Ringle, C.M. and Sarstedt, M. (2015), "A new criterion for assessing discriminant validity in variance-based structural equation modeling", *Journal of the Academy of Marketing Science*, Vol. 43 No. 1, pp. 115-135, doi: [10.1007/s11747-014-0403-8](https://doi.org/10.1007/s11747-014-0403-8).
- Heubeck, T. (2023), "Looking back to look forward: a systematic review of and research agenda for dynamic managerial capabilities", in *Management Review Quarterly*, Springer International Publishing, Issue 0123456789, doi: [10.1007/s11301-023-00359-z](https://doi.org/10.1007/s11301-023-00359-z).
- Horodyski, P. (2023a), "Applicants' perception of artificial intelligence in the recruitment process", *Computers in Human Behavior Reports*, Vol. 11, May, 100303, doi: [10.1016/j.chbr.2023.100303](https://doi.org/10.1016/j.chbr.2023.100303).
- Horodyski, P. (2023b), "Recruiter's perception of artificial intelligence (AI)-based tools in recruitment", *Computers in Human Behavior Reports*, Vol. 10, May, 100298, doi: [10.1016/j.chbr.2023.100298](https://doi.org/10.1016/j.chbr.2023.100298).

- Jatobá, M.N., Ferreira, J.J., Fernandes, P.O. and Teixeira, J.P. (2023), "Intelligent human resources for the adoption of artificial intelligence: a systematic literature review", *Journal of Organizational Change Management*, Vol. 36 No. 7, pp. 1099-1124, doi: [10.1108/JOCM-03-2022-0075](https://doi.org/10.1108/JOCM-03-2022-0075).
- Jay, B. (1991), "Firm resources and sustained competitive advantage", *Journal of Management*, Vol. 17 No. 1, pp. 99-120, doi: [10.1177/014920639101700108](https://doi.org/10.1177/014920639101700108).
- Jia, H.J., Zhuang, Z.Y., Xie, Y.X., Wang, Y.X. and Wu, S.Y. (2023), "Research on dynamic capability and enterprise open innovation", *Sustainability*, Vol. 15 No. 2, pp. 1-23, doi: [10.3390/su15021234](https://doi.org/10.3390/su15021234).
- Jöhnk, J., Weißert, M. and Wyrski, K. (2021), "Ready or not, AI comes- an interview study of organizational AI readiness factors", *Business and Information Systems Engineering*, Vol. 63 No. 1, pp. 5-20, doi: [10.1007/s12599-020-00676-7](https://doi.org/10.1007/s12599-020-00676-7).
- Kammerer, B. (2022), "Hired by a robot: the legal implications of artificial intelligence video interviews and advocating for greater protection of job applicants", *Iowa Law Review*, Vol. 107 No. 2, pp. 817-849.
- Kaushal, N., Kaurav, R.P.S., Sivathanu, B. and Kaushik, N. (2023), "Artificial intelligence and HRM: identifying future research Agenda using systematic literature review and bibliometric analysis", *Management Review Quarterly*, Vol. 73 No. 2, pp. 455-493, doi: [10.1007/s11301-021-00249-2](https://doi.org/10.1007/s11301-021-00249-2).
- Kelan, E.K. (2023), "Algorithmic inclusion: shaping the predictive algorithms of artificial intelligence in hiring", *Human Resource Management Journal*, Vol. 2020 No. 3, pp. 1-14, doi: [10.1111/1748-8583.12511](https://doi.org/10.1111/1748-8583.12511).
- King, K.G. (2016), "Data analytics in human resources: a case study and critical review", *Human Resource Development Review*, Vol. 15 No. 4, pp. 487-495, doi: [10.1177/1534484316675818](https://doi.org/10.1177/1534484316675818).
- Köchling, A. and Wehner, M.C. (2020), "Discriminated by an algorithm: a systematic review of discrimination and fairness by algorithmic decision-making in the context of HR recruitment and HR development", *Business Research*, Vol. 13 No. 3, pp. 795-848, doi: [10.1007/s40685-020-00134-w](https://doi.org/10.1007/s40685-020-00134-w).
- Köchling, A. and Wehner, M.C. (2023), "Better explaining the benefits why AI? Analyzing the impact of explaining the benefits of AI-supported selection on applicant responses", *International Journal of Selection and Assessment*, Vol. 31 No. 1, pp. 45-62, doi: [10.1111/ijsa.12412](https://doi.org/10.1111/ijsa.12412).
- Krakowski, S., Luger, J. and Raisch, S. (2023), "Artificial intelligence and the changing sources of competitive advantage", *Strategic Management Journal*, Vol. 44 No. 6, pp. 1425-1452, doi: [10.1002/smj.3387](https://doi.org/10.1002/smj.3387).
- Kuzior, A., Sira, M. and Brożek, P. (2023), "Use of artificial intelligence in terms of open innovation process and management", *Sustainability*, Vol. 15 No. 9, p. 7205, doi: [10.3390/su15097205](https://doi.org/10.3390/su15097205).
- Lacroux, A. and Martin, C. (2022), "Should I trust the artificial intelligence to recruit? Recruiters' perceptions and behavior when faced with algorithm-based recommendation systems during resume screening", *Frontiers in Psychology*, Vol. 13, July, 895997, doi: [10.3389/fpsyg.2022.895997](https://doi.org/10.3389/fpsyg.2022.895997).
- Langer, M., König, C.J. and Busch, V. (2021), "Changing the means of managerial work: effects of automated decision support systems on personnel selection tasks", *Journal of Business and Psychology*, Vol. 36 No. 5, pp. 751-769, doi: [10.1007/s10869-020-09711-6](https://doi.org/10.1007/s10869-020-09711-6).
- Lee, C.H. and Cha, K.J. (2023), "FAT-CAT-explainability and augmentation for an AI system: a case study on AI recruitment-system adoption", *International Journal of Human-Computer Studies*, Vol. 171, 102976, November 2022, doi: [10.1016/j.ijhcs.2022.102976](https://doi.org/10.1016/j.ijhcs.2022.102976).
- Liao, S.H., Fei, W.C. and Chen, C.C. (2007), "Knowledge sharing, absorptive capacity, and innovation capability: an empirical study of Taiwan's knowledge-intensive industries", *Journal of Information Science*, Vol. 33 No. 3, pp. 340-359, doi: [10.1177/0165551506070739](https://doi.org/10.1177/0165551506070739).
- Liao, S.H., Chang, W.J., Hu, D.C. and Yueh, Y.L. (2012), "Relationships among organizational culture, knowledge acquisition, organizational learning, and organizational innovation in Taiwan's banking and insurance industries", *International Journal of Human Resource Management*, Vol. 23 No. 1, pp. 52-70, doi: [10.1080/09585192.2011.599947](https://doi.org/10.1080/09585192.2011.599947).

- Long, C.S., Wan Ismail, W.K. and Amin, S.M. (2013), "The role of change agent as mediator in the relationship between HR competencies and organizational performance", *International Journal of Human Resource Management*, Vol. 24 No. 10, pp. 2019-2033, doi: [10.1080/09585192.2012.725080](https://doi.org/10.1080/09585192.2012.725080).
- Majhi, S.G., Mukherjee, A. and Anand, A. (2023), "Role of information technology in enabling managerial dynamic capabilities", *VINE Journal of Information and Knowledge Management Systems*, Vol. 53 No. 1, pp. 187-204, doi: [10.1108/VJIKMS-09-2020-0168](https://doi.org/10.1108/VJIKMS-09-2020-0168).
- Meho, L.I. and Rogers, Y. (2023), "Future of jobs report", *World Economic Forum*, Vol. 59 No. 11, pp. 1711-1726, doi: [10.1002/asi.20874](https://doi.org/10.1002/asi.20874).
- Mendoza-Silva, A. (2020), "Innovation capability: a systematic literature review", *European Journal of Innovation Management*, Vol. 24 No. 3, pp. 707-734, doi: [10.1108/EJIM-09-2019-0263](https://doi.org/10.1108/EJIM-09-2019-0263).
- Meshram, R. (2023), "The role of artificial intelligence (Ai) in recruitment and selection of employees in the organisation", *Russian Law Journal*, Vol. 11 No. 9s, pp. 322-333, doi: [10.52783/rlj.v11i9s.1624](https://doi.org/10.52783/rlj.v11i9s.1624).
- Meske, C. and Junglas, I. (2021), "Investigating the elicitation of employees' support towards digital workplace transformation", *Behaviour and Information Technology*, Vol. 40 No. 11, pp. 1120-1136, doi: [10.1080/0144929X.2020.1742382](https://doi.org/10.1080/0144929X.2020.1742382).
- Mirowska, A. and Mesnet, L. (2022), "Preferring the devil you know: potential applicant reactions to artificial intelligence evaluation of interviews", *Human Resource Management Journal*, Vol. 32 No. 2, pp. 364-383, doi: [10.1111/1748-8583.12393](https://doi.org/10.1111/1748-8583.12393).
- Muazu, U.A. and Abdulmalik, S. (2021), "Information technology capabilities and competitive advantage: a review", *International Journal of Technology and Systems*, Vol. 5 No. 1, pp. 1-14, doi: [10.47604/ijts.1206](https://doi.org/10.47604/ijts.1206).
- Murugesan, U., Subramanian, P., Srivastava, S. and Dwivedi, A. (2023), "A study of artificial intelligence impacts on human resource digitalization in Industry 4.0", *Decision Analytics Journal*, Vol. 7, February, 100249, doi: [10.1016/j.dajour.2023.100249](https://doi.org/10.1016/j.dajour.2023.100249).
- Nankervis, A., Connell, J., Cameron, R., Montague, A. and Prikshat, V. (2021), "'Are we there yet?' Australian HR professionals and the fourth industrial revolution", *Asia Pacific Journal of Human Resources*, Vol. 59 No. 1, pp. 3-19, doi: [10.1111/1744-7941.12245](https://doi.org/10.1111/1744-7941.12245).
- Noble, S.M., Foster, L.L. and Craig, S.B. (2021), "The procedural and interpersonal justice of automated application and resume screening", *International Journal of Selection and Assessment*, Vol. 29 No. 2, pp. 139-153, doi: [10.1111/ijsa.12320](https://doi.org/10.1111/ijsa.12320).
- Obrovčić, T., Vlačić, B. and Dabić, M. (2021), "Open innovation in the manufacturing industry: a review and research agenda", *Technovation*, Vol. 102, 102221, doi: [10.1016/j.technovation.2021.102221](https://doi.org/10.1016/j.technovation.2021.102221).
- Pan, Y., Froese, F., Liu, N., Hu, Y. and Ye, M. (2023), "The adoption of artificial intelligence in employee recruitment: the influence of contextual factors", *International Journal of Human Resource Management*, Vol. 33 No. 6, pp. 1125-1147, doi: [10.1080/09585192.2021.1879206](https://doi.org/10.1080/09585192.2021.1879206).
- Park, H., Kim, S., Choi, M. and Choi, J. (2021), "Human, technical, and organizational drivers affecting sustainability of content firms through management and innovation capability during Covid-19", *Sustainability*, Vol. 13 No. 12, p. 6661, doi: [10.3390/su13126661](https://doi.org/10.3390/su13126661).
- Penrose, E.G. (1959), *The Theory of the Growth of the Firm*, John Wiley & Sons, New York.
- Perifanis, N.A. and Kitsios, F. (2023), "Investigating the influence of artificial intelligence on business value in the digital era of strategy: a literature review", *Information*, Vol. 14 No. 2, p. 85, doi: [10.3390/info14020085](https://doi.org/10.3390/info14020085).
- Peteraf, M., Di Stefano, G. and Verona, G. (2013), "The elephant in the room of dynamic capabilities: bringing two diverging conversations together", *Strategic Management Journal*, Vol. 34 No. 12, pp. 1389-1410, doi: [10.1002/smj.2078](https://doi.org/10.1002/smj.2078).
- Pillai, R. and Sivathanu, B. (2020), "Adoption of artificial intelligence (AI) for talent acquisition in IT/ITeS organizations", *Benchmarking: An International Journal*, Vol. 27 No. 9, pp. 2599-2629, doi: [10.1108/BIJ-04-2020-0186](https://doi.org/10.1108/BIJ-04-2020-0186).

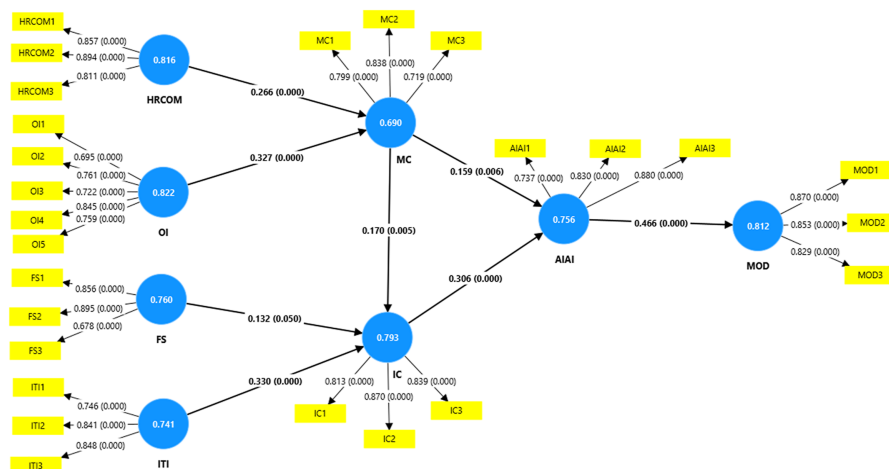
- Prikshat, V., Islam, M., Patel, P., Malik, A., Budhwar, P. and Gupta, S. (2023), "AI-Augmented HRM: literature review and a proposed multilevel framework for future research", *Technological Forecasting and Social Change*, Vol. 193, April, 122645, doi: [10.1016/j.techfore.2023.122645](https://doi.org/10.1016/j.techfore.2023.122645).
- Rahman, M., Ming, T.H., Baigh, T.A. and Sarker, M. (2022), "Adoption of artificial intelligence in banking services: an empirical analysis", *International Journal of Emerging Markets*, Vol. 18 No. 10, pp. 4270-4300, doi: [10.1108/IJOEM-06-2020-0724](https://doi.org/10.1108/IJOEM-06-2020-0724).
- Ratnam, D.S. and Devi, V.R. (2024), "Addressing impediments to HR analytics adoption: guide to HRD professionals", *Human Resource Development International*, Vol. 27 No. 1, pp. 142-151, doi: [10.1080/13678868.2023.2195986](https://doi.org/10.1080/13678868.2023.2195986).
- Rihayana, I.G., Supartha, W.G., Sintaasih, D.K. and Surya, I.B.K. (2023), "Rethinking open innovation from resources based view and dynamic capability perspective: determinants and its consequences", *Quality - Access to Success*, Vol. 24 No. 192, pp. 148-153, doi: [10.47750/QAS/24.192.17](https://doi.org/10.47750/QAS/24.192.17).
- Saunila, M. and Ukko, J. (2014), "Intangible aspects of innovation capability in SMEs: impacts of size and industry", *Journal of Engineering and Technology Management - JET-M*, Vol. 33, pp. 32-46, doi: [10.1016/j.jengtecman.2014.02.002](https://doi.org/10.1016/j.jengtecman.2014.02.002).
- Saura, J.R., Skare, M. and Riberio-Navarret, S. (2022), "How does technology enable competitive advantage? Reviewing state of the art and outlining future directions", *Journal of Competitiveness*, Vol. 14 No. 4, pp. 172-188, doi: [10.7441/joc.2022.04.10](https://doi.org/10.7441/joc.2022.04.10).
- Shet, S.V. and Pereira, V. (2021), "Proposed managerial competencies for Industry 4.0 – implications for social sustainability", *Technological Forecasting and Social Change*, Vol. 173, March, 121080, doi: [10.1016/j.techfore.2021.121080](https://doi.org/10.1016/j.techfore.2021.121080).
- Shet, S.V., Poddar, T., Wamba Samuel, F. and Dwivedi, Y.K. (2021), "Examining the determinants of successful adoption of data analytics in human resource management – a framework for implications", *Journal of Business Research*, Vol. 131, August 2020, pp. 311-326, doi: [10.1016/j.jbusres.2021.03.054](https://doi.org/10.1016/j.jbusres.2021.03.054).
- Su, Y.S., Suen, H.Y. and Hung, K.E. (2021), "Predicting behavioral competencies automatically from facial expressions in real-time video-recorded interviews", *Journal of Real-Time Image Processing*, Vol. 18 No. 4, pp. 1011-1021, doi: [10.1007/s11554-021-01071-5](https://doi.org/10.1007/s11554-021-01071-5).
- Suen, H.Y. and Hung, K.E. (2023), "Building trust in automatic video interviews using various AI interfaces: tangibility, immediacy, and transparency", *Computers in Human Behavior*, Vol. 143, February, 107713, doi: [10.1016/j.chb.2023.107713](https://doi.org/10.1016/j.chb.2023.107713).
- Teece, D.J., Pisano, G. and Shuen, A. (1997), "Dynamic capabilities and strategic management", *Strategic Management Journal*, Vol. 18, March, pp. 77-116, doi: [10.1093/oso/9780198781806.003.0019](https://doi.org/10.1093/oso/9780198781806.003.0019).
- Van den Broek, E., Sergeeva, A. and Huysman, M. (2021), "When the machine meets the expert: an ethnography of developing AI for hiring", *MIS Quarterly: Management Information Systems*, Vol. 45 No. 3, pp. 1557-1580, doi: [10.25300/MISQ/2021/16559](https://doi.org/10.25300/MISQ/2021/16559).
- Van Esch, P. and Black, J.S. (2019), "Marketing AI recruitment: the next phase in job application and selection", *Computers in Human Behavior*, Vol. 90, April 2018, pp. 215-222, doi: [10.1016/j.chb.2018.09.009](https://doi.org/10.1016/j.chb.2018.09.009).
- Voss, G.B., Sirdeshmukh, D. and Voss, Z.G. (2008), "The effects of slack resources and environmental threat on product exploration and exploitation", *Academy of Management Journal*, Vol. 51 No. 1, pp. 147-164, doi: [10.5465/AMJ.2008.30767373](https://doi.org/10.5465/AMJ.2008.30767373).
- Wang, Z., Wang, Q., Zhao, X., Lyles, M.A. and Zhu, G. (2016), "Interactive effects of external knowledge sources and internal resources on the innovation capability of Chinese manufacturers", *Industrial Management and Data Systems*, Vol. 116 No. 8, pp. 1617-1635, doi: [10.1108/IMDS-10-2015-0412](https://doi.org/10.1108/IMDS-10-2015-0412).
- Weber, M., Engert, M., Schaffer, N., Weking, J. and Krcmar, H. (2023), "Organizational capabilities for AI implementation-coping with inscrutability and data dependency in AI", *Information Systems Frontiers*, Vol. 25 No. 4, pp. 1549-1569, doi: [10.1007/s10796-022-10297-y](https://doi.org/10.1007/s10796-022-10297-y).
- Wonglimpiyarat, J. (2010), "Innovation index and the innovative capacity of nations", *Futures*, Vol. 42 No. 3, pp. 247-253, doi: [10.1016/j.futures.2009.11.010](https://doi.org/10.1016/j.futures.2009.11.010).

Appendix 1

Table A1. Constructs, indicators and sources

Constructs	Items	Source
Competency	HRC1. I would feel very competent when I am used to technology like AI-based recruitment HRC2. I get a chance to enhance my capability in many ways by using technology like AI-based recruitment HRC3. I have mastered the ability to use technologies such as AI-based recruitment	Deci <i>et al.</i> (1991), Meske and Junglas (2021)
Open innovation	OI1. External partners, such as vendors, AI Providers, research institutes, consultants, or suppliers are directly involved in all our AI recruitment OI2. Our AI recruitment is highly dependent upon the contribution of external partners, such as vendors, AI Providers, research institutes, consultants, or suppliers OI3. Our firm often buys R&D-related services from external partners, such as vendors, AI Providers, research institutes, consultants, or suppliers OI4. Our firm often buys intellectual property, such as patents, copyrights, or trademarks, from external partners to be used in our AI recruitment OI5. Our firm invests in other firms because we would like to obtain synergies that are beneficial to our innovation recruitment	Cheng and Huizingh (2014)
Financial support	FS1. Our organization fully supports all expenses necessary for the development of AI-based recruitment technologies FS2. Our organization allocates a new budget annually for AI adoption and related technological advancements FS3. Our organization can readily secure financial support from local banks or other financial institutions for AI technology implementation in recruitment processes	Park <i>et al.</i> (2021)
IT infrastructure organizational	IT1. The firm needs a lot of product/service related information for customization IT2. Organization requires significant data and information to customize AI-based recruitment tools and services IT3. The research and development requirements for integrating AI into our recruitment processes are extensive	Bhatt (2000), Park <i>et al.</i> (2021)
Dynamic managerial capability	DMC1. We can plan and implement new technologies like automated interviews in our organization's setup DMC2. We have technical competence for the planning and development of new technologies like automated interviews in our organization's setup DMC3. We can manage the processes of new technologies like automated interviews in our organization's setup	Dwivedi <i>et al.</i> (2023), Park <i>et al.</i> (2021)
AI adoption	AIA1. AI adoption encourages the integration of automated interviews with other functions AIA2. HR can be involved in strategic-level planning after adopting AI-like automated interviews AIA3. AI improves HR functions in the organization	Agarwal (2023), Pan <i>et al.</i> (2023)
Competitive advantage	CA1: Our firm can catch up with advanced firms by absorbing new technology CA2: Our firm improved its market position by absorbing advanced AI Recruitment CA3: Our firm attained cost advantage by creatively assimilating advanced process technology	Ferreira <i>et al.</i> (2021)

Appendix 2

Figure A1. The model path coefficient, factor loading and *p*-value

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